

Some recap from last week...

- We can build *distributional* models of word meaning
- The result is a geometric space of word meaning
- This gives us a metric for word similarity, relatedness, or other relational meaning.
- Reviewing and completing the learning and properties of word embeddings:

Approach: predict if candidate word c is a "neighbor"

1. Treat the target word t and a neighboring context word c as **positive examples**.
2. Randomly sample other words in the lexicon to get negative examples
3. Use logistic regression to train a classifier to distinguish those two cases
4. Use the learned weights as the embeddings

Skip-Gram Training Data

Assume a ± 2 word window, given training sentence:

...lemon, a [tablespoon of apricot jam, a] pinch...

c1 c2 [target] c3 c4

Skip-Gram Classifier

(assuming a +/- 2 word window)

...lemon, a [tablespoon of apricot jam, a] pinch...
 c1 c2 [target] c3 c4

Goal: train a classifier that is given a candidate (**w**ord, **c**ontext) pair

(apricot, jam)

(apricot, aardvark)

...

And assigns each pair a probability:

$$P(+ | w, c)$$

$$P(- | w, c) = 1 - P(+ | w, c)$$

Skip-Gram Training data

...lemon, a [tablespoon of apricot jam, a] pinch...

c1

c2

[target]

c3

c4



positive examples +

t

c

apricot tablespoon

apricot of

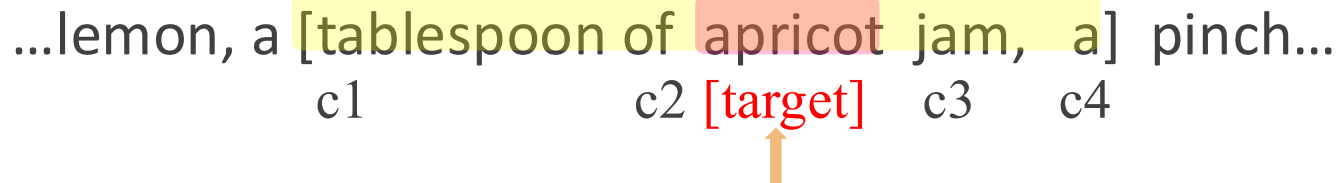
apricot jam

apricot a

Skip-Gram Training data

...lemon, a [tablespoon of apricot jam, a] pinch...

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positive examples +

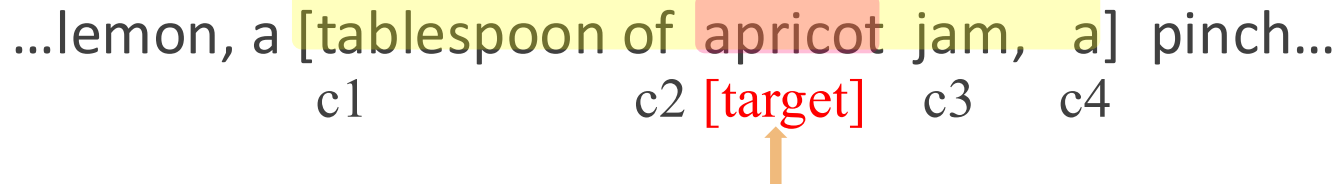
t	c
apricot	tablespoon
apricot	of
apricot	jam
apricot	a

For each positive example we'll grab k negative examples, sampling by frequency

Skip-Gram Training data

...lemon, a [tablespoon of apricot jam, a] pinch...

c1 c2 [target] c3 c4



positive examples +

t	c
apricot	tablespoon
apricot	of
apricot	jam
apricot	a

negative examples -

t	c	t	c
apricot	aardvark	apricot	seven
apricot	my	apricot	forever
apricot	where	apricot	dear
apricot	coaxial	apricot	if

The kinds of neighbors depend on window size

Small windows ($C = \pm 2$) : nearest words are syntactically similar words in same taxonomy

- *Hogwarts* nearest neighbors are other fictional schools
- *Sunnydale, Evernight, Blandings*

Large windows ($C = \pm 5$) : nearest words are related words in same semantic field

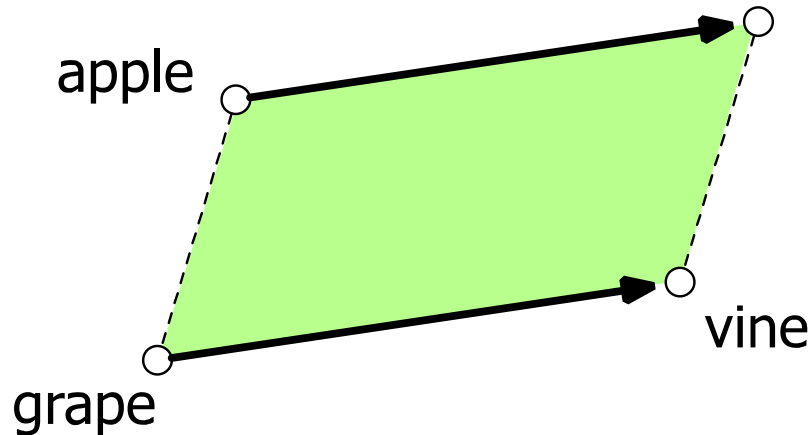
- *Hogwarts* nearest neighbors are Harry Potter world:
- *Dumbledore, half-blood, Malfoy*

Analogical relations

The classic parallelogram model of analogical reasoning
(Rumelhart and Abrahamson 1973)

To solve: *"apple is to tree as grape is to _____"*

*Add $\overrightarrow{\text{tree} - \text{apple}}$ to $\overrightarrow{\text{grape}}$ to get **vine***



Analogical relations via parallelogram

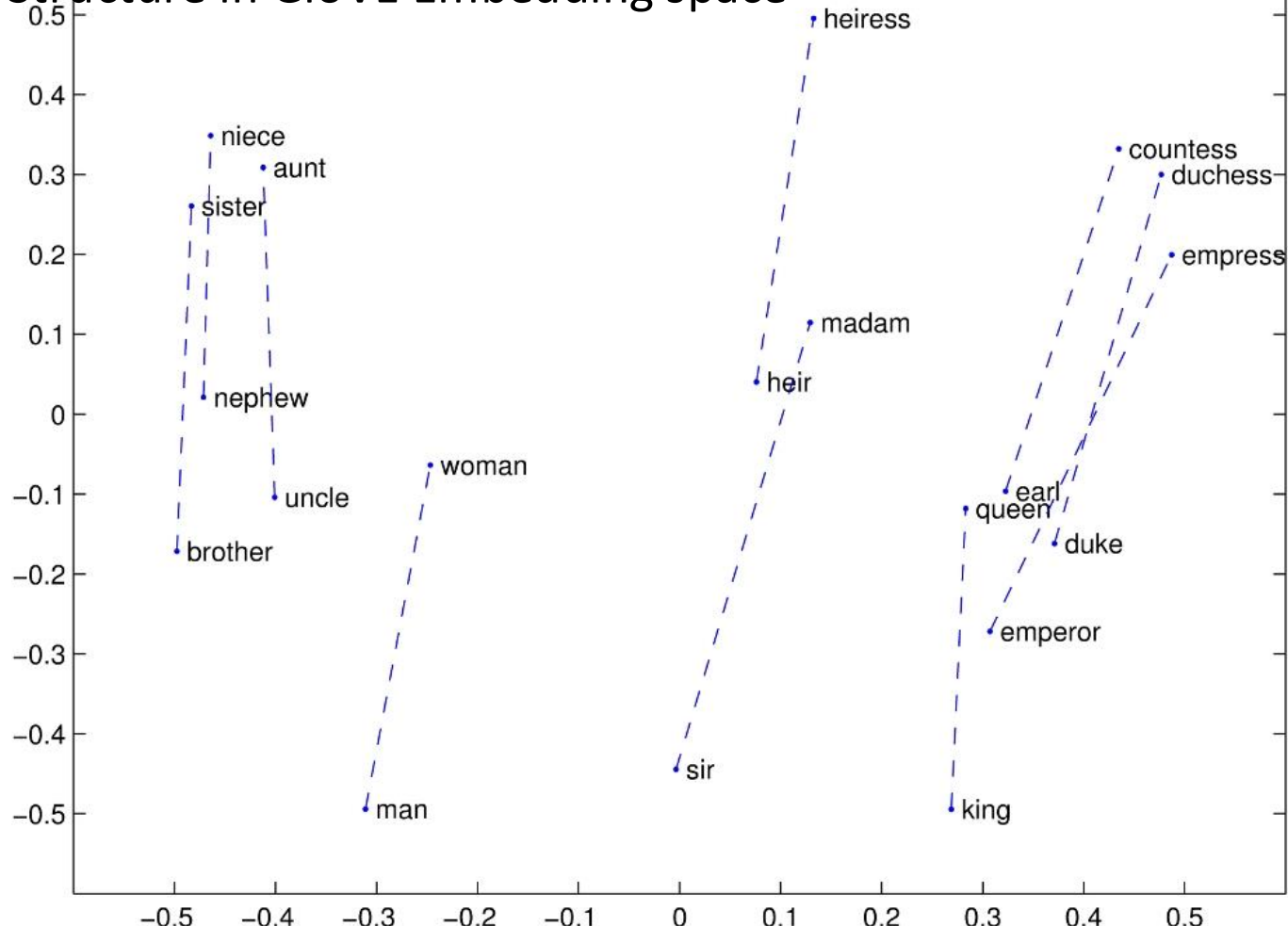
The parallelogram method can solve analogies with both sparse and dense embeddings (Turney and Littman 2005, Mikolov et al. 2013b)

$\overrightarrow{\text{king}} - \overrightarrow{\text{man}} + \overrightarrow{\text{woman}}$ is close to $\overrightarrow{\text{queen}}$
 $\overrightarrow{\text{Paris}} - \overrightarrow{\text{France}} + \overrightarrow{\text{Italy}}$ is close to $\overrightarrow{\text{Rome}}$

For a problem $a:a^*:b:b^*$, the parallelogram method is:

$$\hat{b}^* = \operatorname{argmax}_x \text{distance}(x, a^* - a + b)$$

Structure in GloVe Embedding space



Caveats with the parallelogram method

It only seems to work for frequent words, small distances and certain relations (relating countries to capitals, or parts of speech), but not others. (Linzen 2016, Gladkova et al. 2016, Ethayarajh et al. 2019a)

Understanding analogy is an open area of research (Peterson et al. 2020)

Embeddings reflect cultural bias!

Bolukbasi, Tolga, Kai-Wei Chang, James Y. Zou, Venkatesh Saligrama, and Adam T. Kalai. "Man is to computer programmer as woman is to homemaker? debiasing word embeddings." In *NeurIPS*, pp. 4349-4357. 2016.

Ask “Paris : France :: Tokyo : x”

- x = Japan

Ask “father : doctor :: mother : x”

- x = nurse

Ask “man : computer programmer :: woman : x”

- x = homemaker

Algorithms that use embeddings as part of e.g., hiring searches for programmers, might lead to bias in hiring

Sentences

Natural Language Processing, Week 3, Summer 2026

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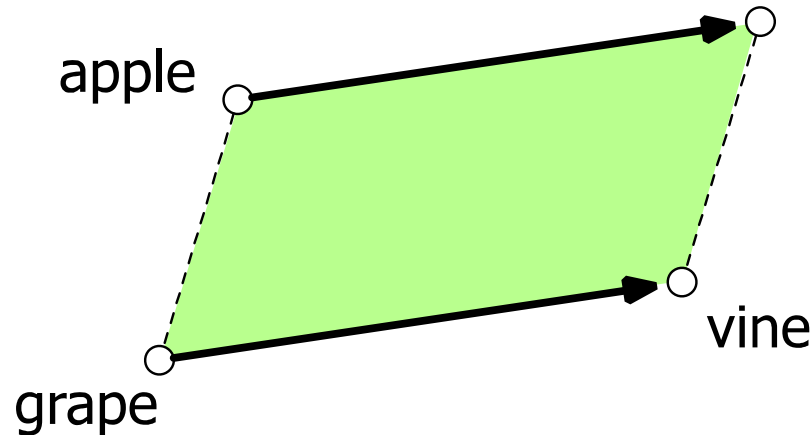
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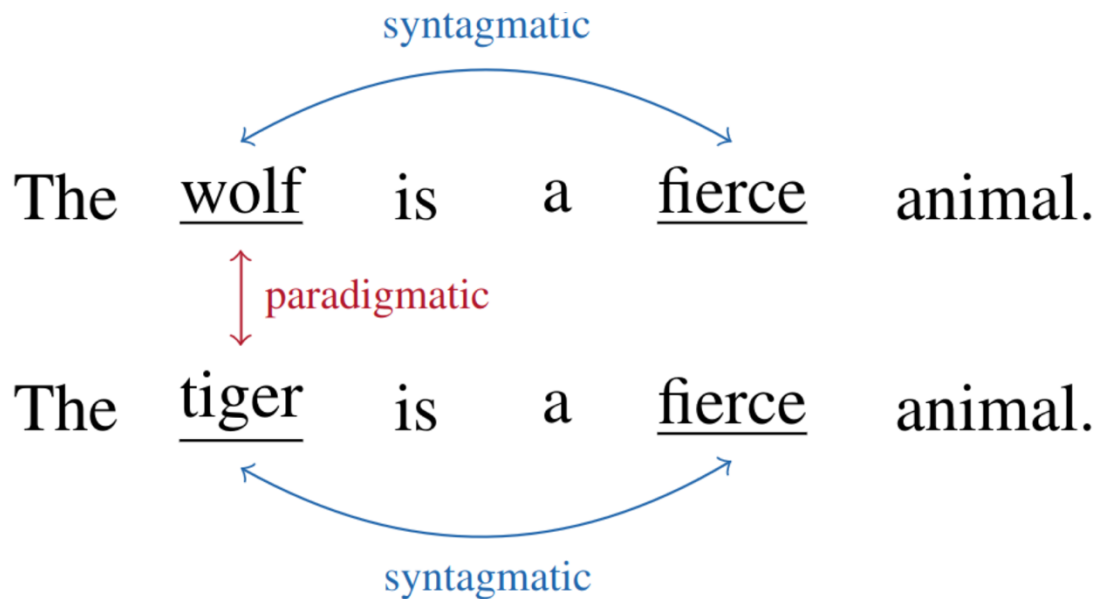
To solve: "*apple is to tree as grape is to _____*"

*Add tree – apple to grape to get **vine***



Distributional measures of conceptual association

- Words that are conceptually **related** tend to **occur together** in the same context: “related” and “occur together” are richly ambiguous and vague here



- (example from Sun, Fei, et al. "Learning word representations by jointly modeling syntagmatic and paradigmatic relations." 2015)

Co-occurrence context windows

- Contiguous windows (or samples from windows) of 2-5 words around focal token
- Natural boundaries in text: sentence, paragraph, document
- Syntactic or relational context (e.g. x occurs as adjectival modifier of y)
- Sliding window at a distance from the focal token
- Specific relational contexts indicated structured relations, e.g. “Xs such as Y” indicates Y is type of X. (hyponym)¹

1. Hearst, Marti A. "Automatic acquisition of hyponyms from large text corpora." *Proceedings of the 14th conference on Computational linguistics-Volume 2*. Association for Computational Linguistics, 1992.

Corpus linguistics and word meaning

Word embeddings can be opaque and difficult to interpret

We can still use computers to assist more traditional corpus linguistics or stylistics work.

.. and in addition to lexical ambiguity, there are other kinds of ambiguity

Keyword in context

freedom from **violence** allows a girl

freedom from **violence** is so fundam

freedom from **risk** of disease. </s><

freedom from foreign **rule** and its bi

freedom from unwarranted **searche**

freedom from **violence** and poverty.

freedom from the **pain** they have en

freedom from **imprisonment** . </s>

freedom from **torture** of various Am

freedom from sexual **exploitation** :

freedom from **violence** . </s><s> Ir

freedom from **want** " one of the "fol

Freedom to **Marry** 2015). </s>

Freedom to **Export** to Cuba Ac

Freedom to **Export** to Cuba Ac

Freedom to **Travel** to Cuba Ac

freedoms to **users**). </s><s>

freedom to these endangered

Freedom to **Trade** , Free Trade

freedom to express **ideas** has

freedom to **people** like Washi

Freedom to **Marry** " operation

Keyword in context

his Army are allowed to return to great Britain, on parole not to	bear	Arms against America during the present Contest. The Incendiary
ed to their rightful Sovereign; and rather than renounce Him, or	bear	Arms against Him, have submitted to Death. Unsatisfied with thes
gland; do hereby promise on my Word of Honour, that I will not	bear	Arms against the United States of America, nor do or say any thing
phia was tender'd to me & I had read as far as "That I would not	bear	Arms against the united Colonies for twelve Months[*]; I refus'd to
nstrained our fellow Citizens taken Captive on the high Seas to	bear	Arms against their Country , to become the executioners of their fr
nstrained our fellow Citizens, taken Captive on the high Seas, to	bear	Arms against their Country, to become the Executioners of their Fr
Protection of the Crown, and even compelling some of them to	bear	Arms against their Countrymen: The Treaties of the King of Great-B
nstrained our fellow Citizens taken Captive on the high Seas to	bear	Arms against their Country, to become the executioners of their fri
can you be in any Kingdom or Government, seeing you will not	bear	Arms and fight? To this I replied, that many of us had borne Arms
y with truth all Weymouth Braintree Hingham who were able to	bear	Arms, and hundreds from other Towns within 20 30 and 40 miles
nover and Henrico to assemble immediately every Man able to	bear	Arms, and one half of those of Amelia and Cumberland and to brin
e, that Numbers of the Inhabitants murmur at being Obligated to	bear	Arms; and the dread of a French War is very General: all Accounts
made of all the Men in the several Districts in the State able to	bear	Arms; and will meet on the 16th instant, to take the most effectual
ps of the Colony, filled up with every Canadian that was able to	bear	Arms, besides several Nations of Savages, had taken the Field in a
one to our Establishment; by making it universally reputable to	bear	Arms & disgraceful to decline having a share in the performance o
it hath to require the* personal Service of every ManA able to	bear	Arms in case of a sudden Invasion or formidable Insurrection. The
in the words " vagabonds and " add " all persons who refuse to	bear	Arms in defence of the State to which they belong , and all person

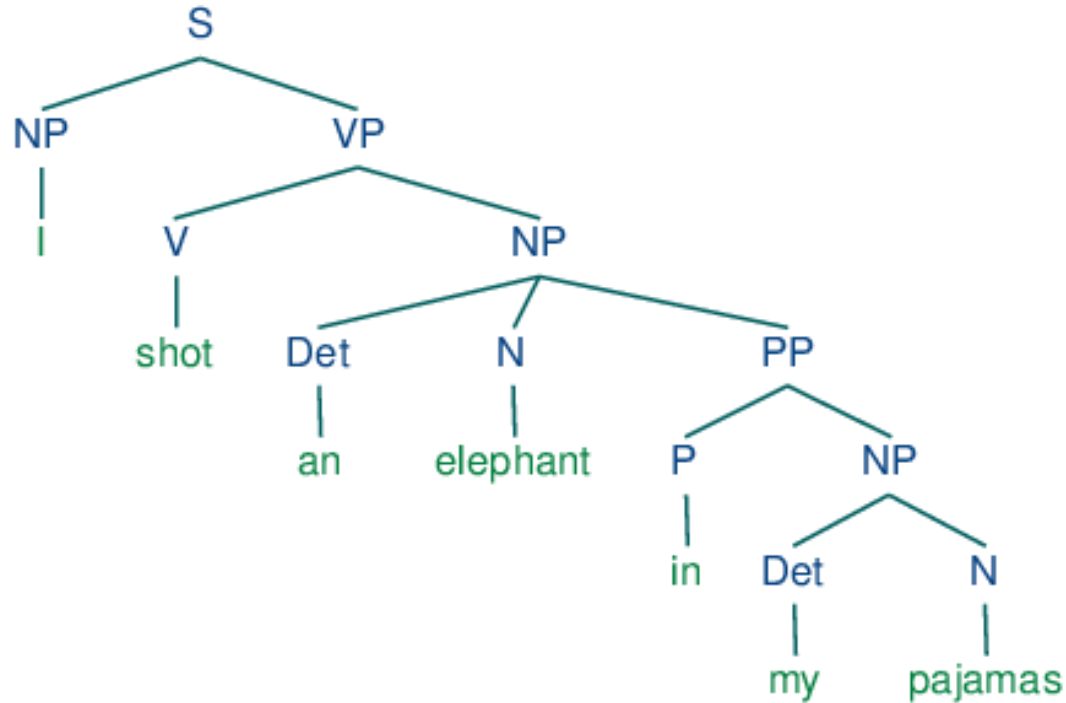
Syntactic and Lexical ambiguity

A well regulated Militia, being necessary to the security of a free State, the right of the people to keep and bear Arms, shall not be infringed.

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Parsing Natural Language



“One morning I shot an elephant in my pajamas. How he got into my pajamas I'll never know”. - Groucho Marx

Controlling for baseline word frequency

- If we measure word association (or similarity?) simply by counts of co-occurrences, words with very frequent overall occurrence rates will dominate

Focal token?

experiment

Year range

1,701 1,780 1,800

1,701 1,711 1,721 1,731 1,741 1,751 1,761 1,771 1,781 1,791 1,800

Distances:

☐ 5

☐ 10

☐ 20

☐ 30

☐ 40

☐ 50

☐ 60

☐ 70

☒ 80

☐ 90

☐ 100

Search!

Show 25 entries

orth	distance	count	boundBase	dpr
the	80	68981	211591169	1140.4042
of	80	42777	134748808	1005.5375
and	80	24390	104329434	699.9578
to	80	23473	93375664	734.5201
in	80	19537	62436906	836.8180
a	80	19506	70504586	759.9343
it	80	11682	30258466	880.3878
that	80	11067	36618221	718.7262
is	80	9697	32056543	698.6180
be	80	8232	25957201	699.2003
i	80	8073	44392600	451.1803
by	80	7816	23602787	714.9745
which	80	7514	19871195	786.0929
with	80	7391	23078380	688.0507

and/or	8,263	0.23	pp obj of	7,638	0.22	pp of	4,087	0.12	pp obj at	2,979	0.08	pp obj to	1,429	0.04	pp obj for	1,081	0.03	pp to	1,055	0.03
property +	664	10.03	friend +	167	8.24	prefs +	264	10.80	let +	399	11.10	dangerous +	108	9.54	fruggle	47	10.23	captive	48	10.20
privilege +	337	9.67	caufe +	346	8.19	confcience +	383	9.99	leave +	127	9.42	favourable	51	8.81	fruggles	32	9.81	choofe	18	8.45
right +	777	9.42	bleffings	80	8.05	subject +	130	9.02	myself	73	9.39	unfriendly	21	8.68	advocate	39	8.89	chufe	18	8.39
independence +	177	9.18	guardian	70	8.01	press	50	8.57	let	55	9.01	friendly	26	8.58	declaration	16	8.79	luggelt	14	8.20
equality +	120	8.59	enjoyment +	119	7.94	ions	47	8.14	himself	77	8.84	friend +	108	8.50	struggle	14	8.62	dilpofe	24	8.18
happinefs +	176	8.35	prefervation	83	7.79	choice	46	7.69	themselves	41	8.23	attachment	34	8.17	zeal	51	8.36	addrefs	13	7.73
freedom +	146	8.19	lweets	56	7.78	lpeech	52	7.67	be +	1,499	8.19	reftored	30	8.08	security	26	8.22	flaves	8	7.73
religion +	295	7.50	deprive +	116	7.76	lsubjects	45	7.59	set	58	7.99	favorable	13	7.89	contend	42	8.16	diffent	8	7.67
security	97	7.47	principle +	275	7.75	America	91	7.58	more	43	7.76	effential	25	7.85	apologize	14	8.04	request	13	7.64
safety +	119	7.45	love +	308	7.68	country +	332	7.56	foul	15	7.28	hollile	11	7.82	pant	12	8.03	fail	8	7.09
life +	701	7.36	idea +	125	7.48	navigation	26	7.47	walk	13	6.97	danger	25	7.81	fruggling	9	7.96	publiih	13	7.09
conftitution	63	7.35	defence +	174	7.46	mankind	91	7.18	prifoners	12	6.93	right	72	7.78	pallion	18	7.96	lpeak	27	6.73
agency	39	7.20	security	73	7.43	Prefs	16	6.83	party	13	6.87	martyr	11	7.77	effort	9	7.86	obferve	10	6.70
law +	255	7.05	palladium	39	7.35	commerce	24	6.69	ourfelves	11	6.86	inimical	11	7.74	fight	34	7.76	ourfelves	7	6.33
immunity	39	6.89	lofs	89	7.33	conscience	65	6.68	let	11	6.85	reftored	24	7.70	exertion	7	7.56	learch	7	5.95
peace +	278	6.84	abufe	48	7.32	colony	32	6.65	ftill	12	6.64	fatal	20	7.49	regard	13	7.35	anfwer	15	5.63
virtue	88	6.76	ions	66	7.30	citizen	27	6.53	herfelf	10	6.62	entitle	19	7.42	struggle	6	7.35	lin	7	5.60
necellity	44	6.73	charter	42	7.22	confciences	13	6.38	mind	9	6.35	attach	18	7.42	wilh	13	7.31	lay	42	5.58
eltates	37	6.71	land +	124	7.12	filhing	11	6.37	letting	7	6.23	deftuctive	13	7.18	champion	6	7.13	leek	8	5.37
juftice	59	6.60	jealous	46	7.05	individual	18	6.36	hand	10	6.21	formidable	9	7.12	bleed	6	7.13	side	11	5.24

Quantitative measures of style

- Subjective experience of reading literature cannot be replicated by machine
- The experience of difficulty or complexity cannot be directly and reliably shared and communicated
- Specifying algorithmic methods for measuring complexity forces us to state generalisable properties that are associated with complexity in formal, replicable, ways that can be reliably communicated
- In many cases, this is what literary critics who write about difficulty do anyway
- The most common use for stylistics is authorship attribution: we will return to this when we cover document classification

Features associated with difficulty and style

Simple or “common” words, short sentences

“Authors should use common words to say uncommon things.” ...

“Good writing should be governed by the rule that a man can think only one thing clearly at a time; and, therefore, that he should not be expected to think two or even more things in one and the same moment. But this is what is done when a writer breaks up his principal sentence into little pieces, for the purpose of pushing into the gaps thus made two or three other thoughts by way of parenthesis; thereby unnecessarily and wantonly confusing the reader.”

- Schopenhauer, Art of Literature

“Adjectives are the sugar of literature and adverbs the salt” : Henry James

Co-reference ambiguities

(26.40) John telephoned Bill. He lost the laptop.

(26.41) John criticized Bill. He lost the laptop.

These examples differ only in the verb used in the first sentence, yet “he” in (26.40) is typically resolved to John, whereas “he” in (26.41) is resolved to Bill. This may be partly due to the link between implicit causality and saliency: the implicit cause of a “criticizing” event is its object, whereas the implicit cause of a “telephoning” event is its subject. In such verbs, the entity which is the implicit cause may be more salient.

Established ‘readability’ formulae

- Designed with education requirements in mind
- Many, many variants (e.g., see <https://en.wikipedia.org/wiki/Readability>)
- Few variables: sentence length, word length in syllables or characters
- Type-token ratio, corrected in various ways
- List of 3000 ‘difficult’ words
- Calibrated to human judgments to produce “Grade level”

$$0.39 \left(\frac{\text{total words}}{\text{total sentences}} \right) + 11.8 \left(\frac{\text{total syllables}}{\text{total words}} \right) - 15.59$$

Established 'readability' formulae: drawbacks

- Components derived from written orthography may not reflect difficulty of understanding spoken communication
- Components like word length may not be robust/reliable outside of English
- Many do not measure vocabulary difficulty directly at all

Stylistics for literature: Portrait of the Artist

Chapter I

Once upon a time and a very good time it was there was a moocow coming down along the road and this moocow that was coming down along the road met a nicens little boy named baby tuckoo....

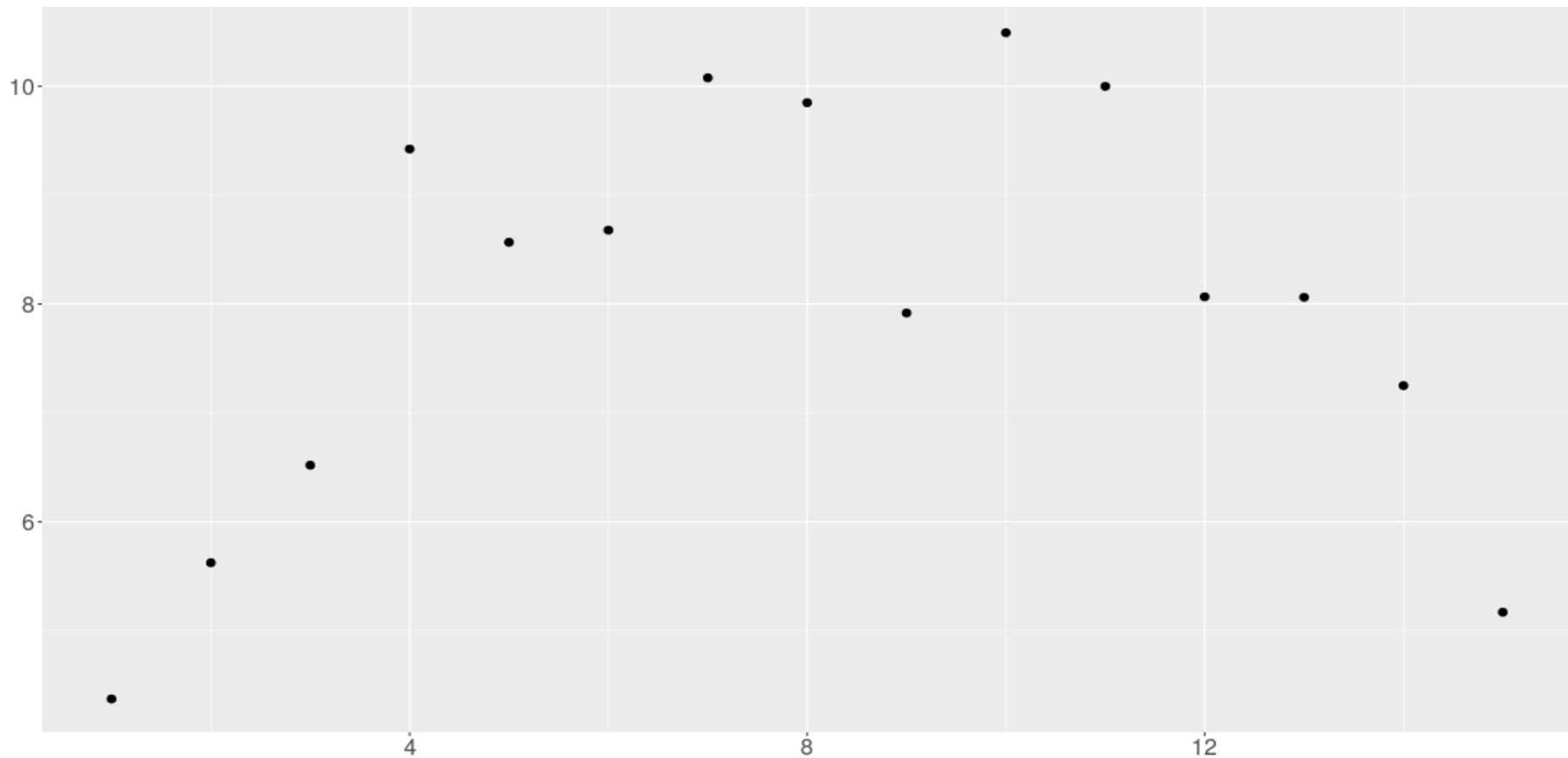
His father told him that story: his father looked at him through a glass: he had a hairy face.

He was baby tuckoo. The moocow came down the road where Betty Byrne lived: she sold lemon platt.

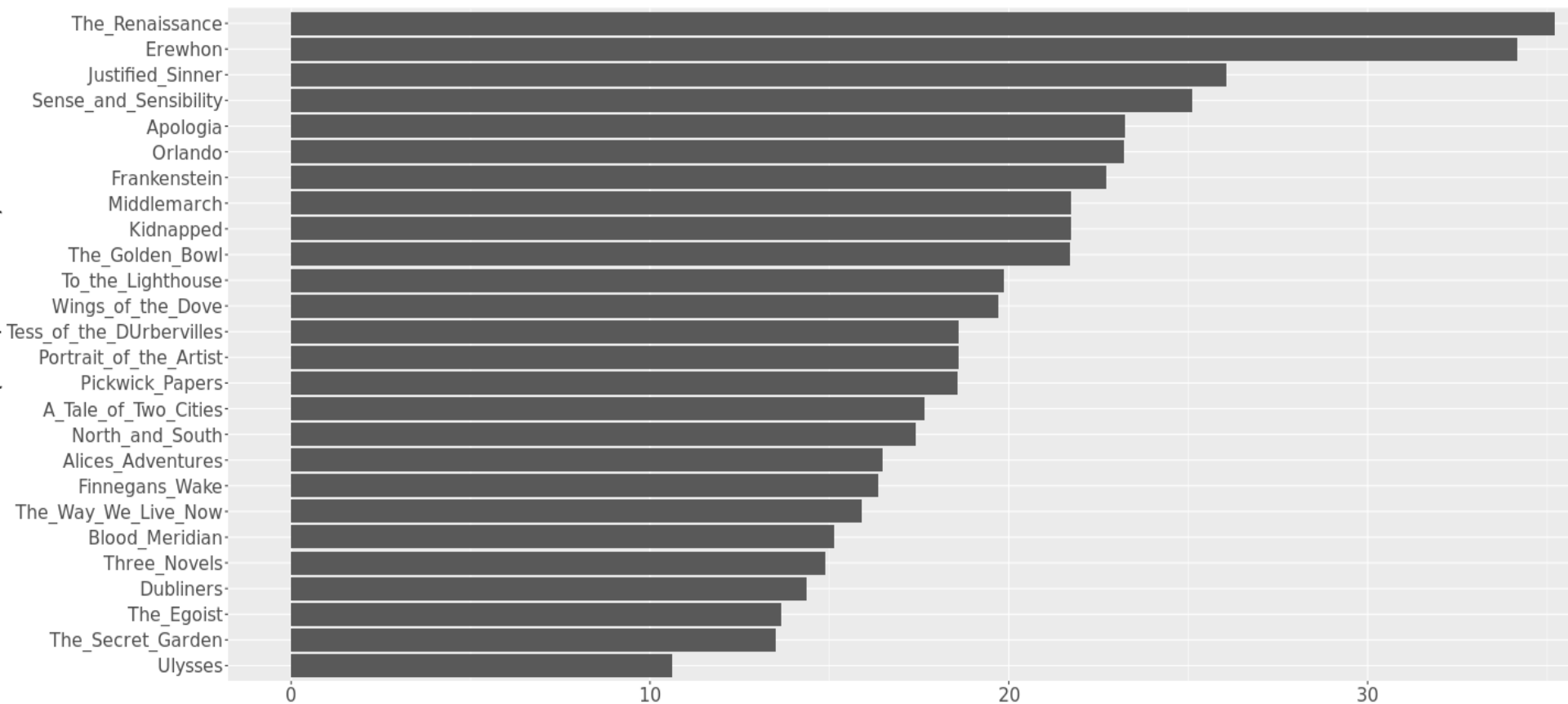
Chapter V

Leaning against the fireplace Stephen heard him greet briskly and impartially every student of the class and could almost see the frank smiles of the coarser students. A desolating pity began to fall like dew upon his easily embittered heart for this faithful servingman of the knightly Loyola, for this halfbrother of the clergy, more venal than they in speech, more steadfast of soul than they, one whom he would never call his ghostly father; and he thought how this man and his companions had earned the name of worldlings at the hands not of the unworldly only but of the worldly also for having pleaded, during all their history, at the bar of God's justice for the souls of the lax and the lukewarm and the prudent.

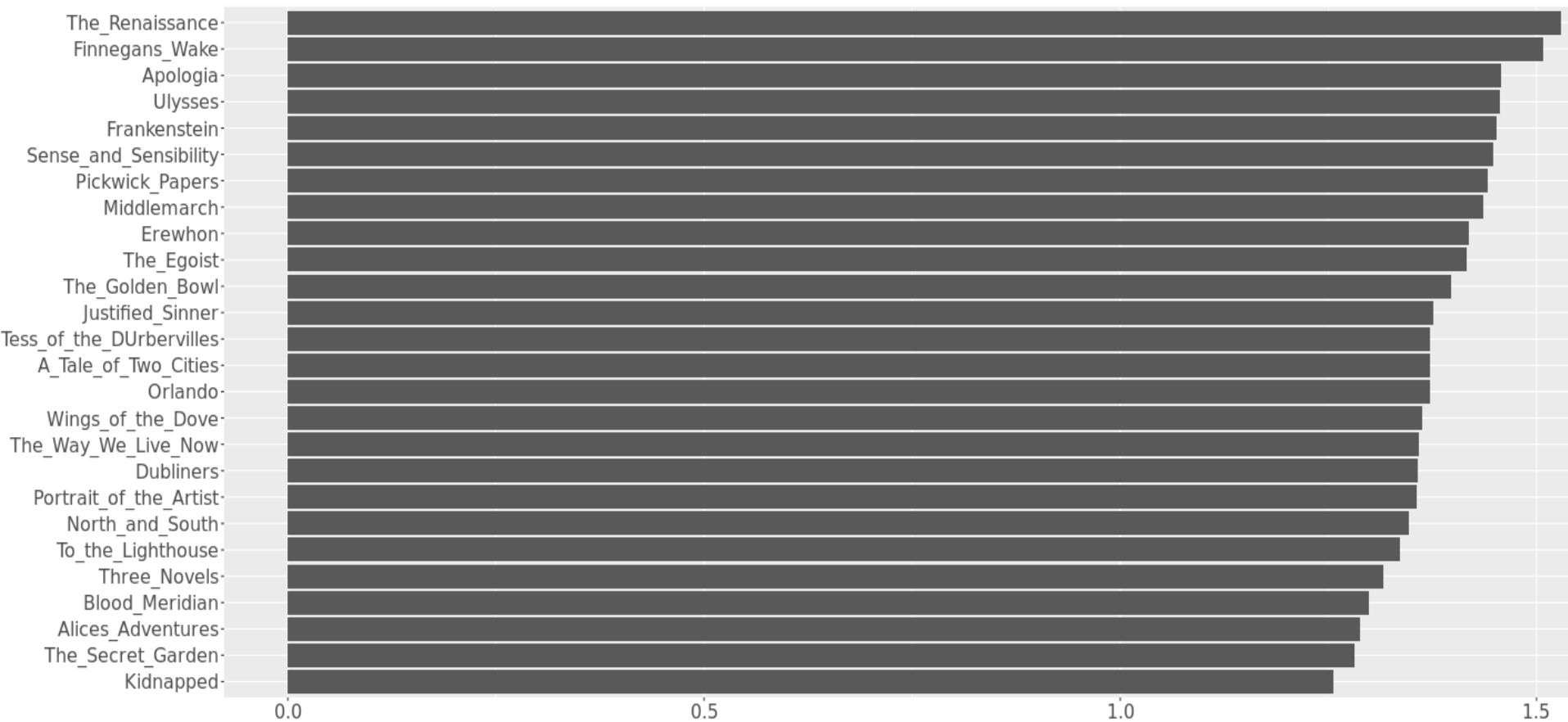
Portrait of the Artist - Flesch-Kincaid



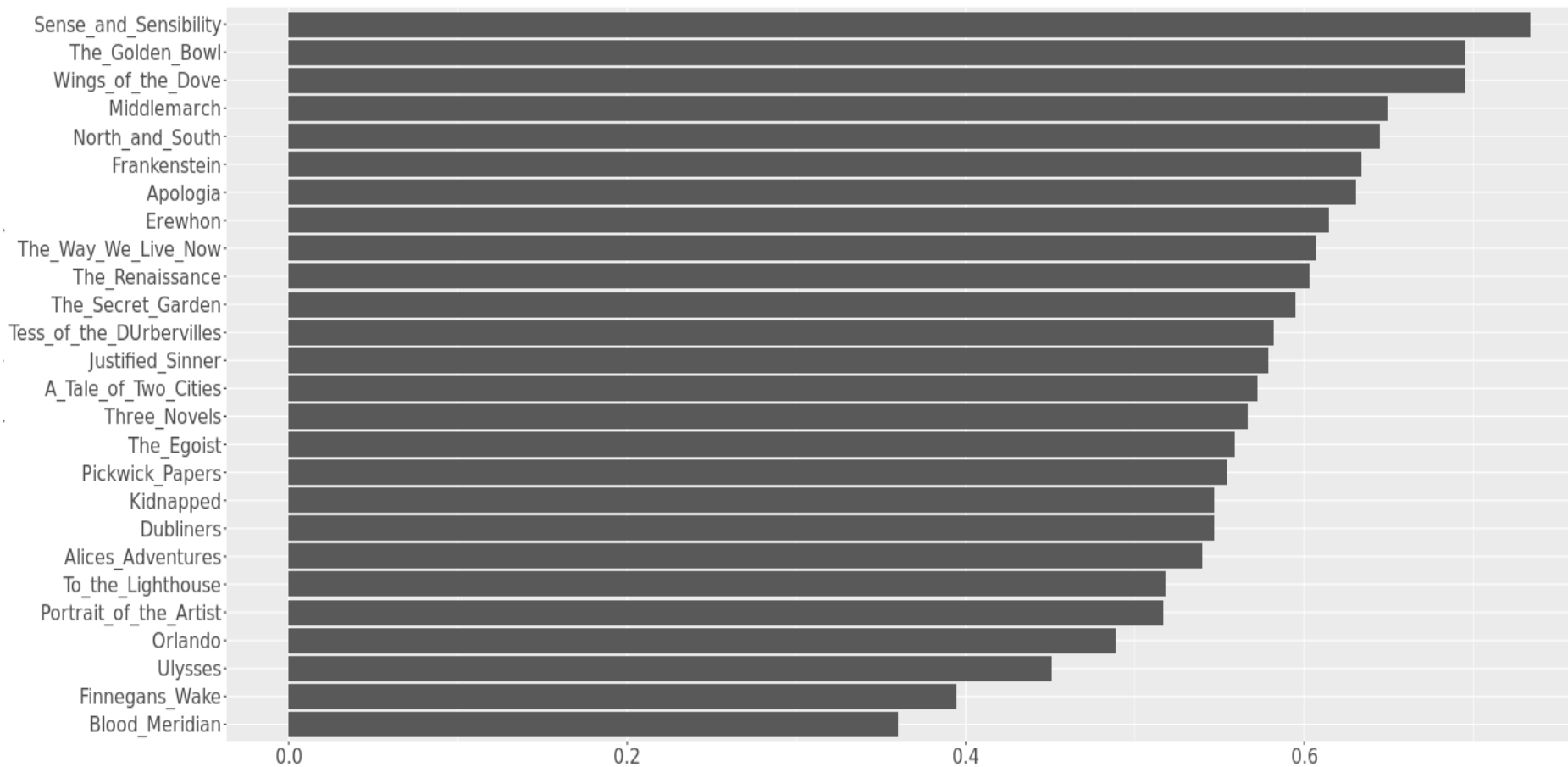
Mean Sentence Length



Mean word syllables



Rate of adjectives per noun



Colorless green ideas: adjective choice

1	-by	grace-
2	decollated	percursor
3	supralapsarian	dogmatists
4	gibbeted	malefactors
5	twined	pierglass
6	tufted	sandknolls
7	crimped	surplices
8	vaporous	cloudlets
9	gaudy	playbill
10	moody	mistrustfulness
11	whited	sepulchres
12	sinful	impenitence
13	undistinguished	dullards
14	portly	bursar
15	stale	shite
16	filthy	cowyard
17	jewelly	slippers
18	angelical	natures
19	angelical	trumpet
20	fouled	wenches

Frequent adjectival choices

- 1 implicit faith
- 2 shrill cries
- 3 workaday world
- 4 shabby clothes
- 5 evil spirits
- 6 broad grin
- 7 universal brotherhood
- 8 impenetrable darkness
- 9 downcast eyes
- 10 parched lips
- 11 heavenly bodies
- 12 elder brother
- 13 hooked nose
- 14 olden times
- 15 eldest son
- 16 nervous system
- 17 shrill whistle
- 18 eternal damnation
- 19 clenched fist
- 20 buttered toast

But how to get these syntactic relations computationally?

- Syntactic parsing
- Dependency parsing
- spaCy in python
- Stanza: python with Java backend) <https://stanfordnlp.github.io/stanza/>

Sequence Labeling for Part of Speech and Named Entities

Slides in this section from Jurafsk and
Martin, Speech and Language
Processing:

<https://web.stanford.edu/~jurafsky/slp>
3/

Part of Speech Tagging

Parts of Speech

From the earliest linguistic traditions (Yaska and Panini 5th C. BCE, Aristotle 4th C. BCE), the idea that words can be classified into grammatical categories

- part of speech, word classes, POS, POS tags

8 parts of speech attributed to Dionysius Thrax of Alexandria (c. 1st C. BCE):

- noun, verb, pronoun, preposition, adverb, conjunction, participle, article
- These categories are relevant for NLP today.

Two classes of words: Open vs. Closed

Closed class words

- Relatively fixed membership
- Usually **function** words: short, frequent words with grammatical function
 - determiners: *a, an, the*
 - prepositions: *on, under, over, near, by, ...*

Open class words

- Usually **content** words: Nouns, Verbs, Adjectives, Adverbs
 - Plus interjections: *oh, ouch, uh-huh, yes, hello*
- New nouns and verbs like *iPhone* or *to fax*

Open class ("content") words

Nouns

Proper

Janet
Italy

Common

cat, cats
mango

Verbs

Main

eat
went

Auxiliary

can
had

Adjectives

old green tasty

Adverbs

slowly yesterday

Numbers

122,312
one

Interjections *Ow hello*

... more

Closed class ("function")

Determiners *the some*

Conjunctions *and or*

Pronouns *they its*

Prepositions *to with*

Particles *off up*

... more

Part-of-Speech Tagging

Assigning a part-of-speech to each word in a text.

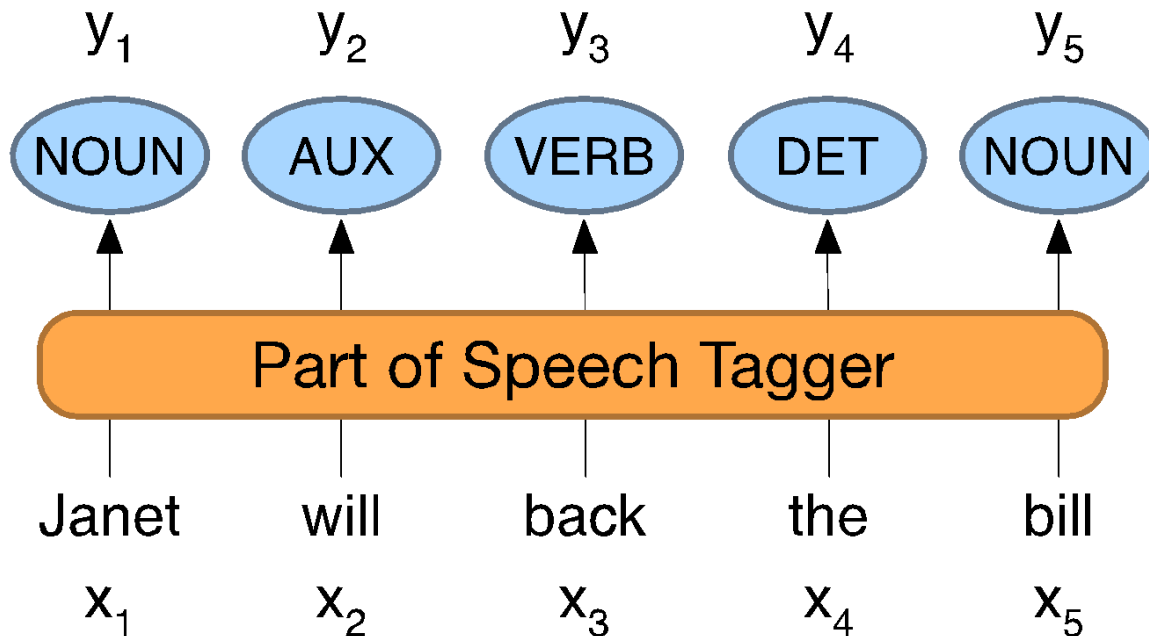
Words often have more than one POS.

book:

- VERB: (***Book** that flight*)
- NOUN: (*Hand me that **book***).

Part-of-Speech Tagging

Map from sequence $x_1, \dots, x_n$ of words to $y_1, \dots, y_n$ of POS tags



"Universal Dependencies" Tagset

Nivre et al.
2016

	Tag	Description	Example
Open Class	ADJ	Adjective: noun modifiers describing properties	<i>red, young, awesome</i>
	ADV	Adverb: verb modifiers of time, place, manner	<i>very, slowly, home, yesterday</i>
	NOUN	words for persons, places, things, etc.	<i>algorithm, cat, mango, beauty</i>
	VERB	words for actions and processes	<i>draw, provide, go</i>
	PROPN	Proper noun: name of a person, organization, place, etc..	<i>Regina, IBM, Colorado</i>
	INTJ	Interjection: exclamation, greeting, yes/no response, etc.	<i>oh, um, yes, hello</i>
Closed Class Words	ADP	Adposition (Preposition/Postposition): marks a noun's spacial, temporal, or other relation	<i>in, on, by under</i>
	AUX	Auxiliary: helping verb marking tense, aspect, mood, etc.,	<i>can, may, should, are</i>
	CCONJ	Coordinating Conjunction: joins two phrases/clauses	<i>and, or, but</i>
	DET	Determiner: marks noun phrase properties	<i>a, an, the, this</i>
	NUM	Numeral	<i>one, two, first, second</i>
	PART	Particle: a preposition-like form used together with a verb	<i>up, down, on, off, in, out, at, by</i>
	PRON	Pronoun: a shorthand for referring to an entity or event	<i>she, who, I, others</i>
Other	SCONJ	Subordinating Conjunction: joins a main clause with a subordinate clause such as a sentential complement	<i>that, which</i>
	PUNCT	Punctuation	<i>; , ()</i>
	SYM	Symbols like \$ or emoji	<i>\$, %</i>
	X	Other	<i>asdf, qwfg</i>

Sample "Tagged" English sentences

There/**PRO** were/**VERB** 70/**NUM** children/**NOUN**
there/**ADV** ./**PUNC**

Preliminary/**ADJ** findings/**NOUN** were/**AUX**
reported/**VERB** in/**ADP** today/**NOUN** 's/**PART**
New/**PROPN** England/**PROPN** Journal/**PROPN**
of/**ADP** Medicine/**PROPN**

Why Part of Speech Tagging?

- Can be useful for other NLP tasks
 - Parsing: POS tagging can improve syntactic parsing
 - MT: reordering of adjectives and nouns (say from Spanish to English)
 - Sentiment or affective tasks: may want to distinguish adjectives or other POS
 - Text-to-speech (how do we pronounce “lead” or “object”?)
- Or linguistic or language-analytic computational tasks
 - Need to control for POS when studying linguistic change like creation of new words, or meaning shift
 - Or control for POS in measuring meaning similarity or difference

How difficult is POS tagging in English?

Roughly 15% of word types are ambiguous

- Hence 85% of word types are unambiguous
- *Janet* is always PROPN, *hesitantly* is always ADV

But those 15% tend to be very common.

So ~60% of word tokens are ambiguous

E.g., *back*

earnings growth took a *back*/ADJ seat

a small building in the *back*/NOUN

a clear majority of senators *back*/VERB the bill

enable the country to buy *back*/PART debt

I was twenty-one *back*/ADV then

POS tagging performance in English

How many tags are correct? (Tag accuracy)

- About 97%
- Hasn't changed in the last 10+ years
- HMMs, CRFs, BERT perform similarly .
- Human accuracy about the same

But baseline is 92%!

- Baseline is performance of stupidest possible method
- "Most frequent class baseline" is an important baseline for many tasks
 - Tag every word with its most frequent tag
 - (and tag unknown words as nouns)
- Partly easy because
 - Many words are unambiguous

Sources of information for POS tagging

Janet will back the bill
AUX/NOUN/VERB? NOUN/VERB?

Prior probabilities of word/tag

- "will" is usually an AUX

Identity of neighboring words

- "the" means the next word is probably not a verb

Morphology and wordshape:

- | | | |
|------------------|--------------|------------|
| ◦ Prefixes | unable: | un- → ADJ |
| ◦ Suffixes | importantly: | -ly → ADJ |
| ◦ Capitalization | Janet: | CAP → PROP |

Standard algorithms for POS tagging

Supervised Machine Learning Algorithms:

- Hidden Markov Models
- Conditional Random Fields (CRF)/ Maximum Entropy Markov Models (MEMM)
- Neural sequence models (RNNs or Transformers)
- Large Language Models (like BERT), finetuned

All required a hand-labeled training set, all about equal performance (97% on English)

All make use of information sources we discussed

- Via human created features: HMMs and CRFs
- Via representation learning: Neural LMs

Sequence
Labeling for
Part of Speech
and Named
Entities

Named Entity Recognition
(NER)

Named Entities

- **Named entity**, in its core usage, means anything that can be referred to with a proper name. Most common 4 tags:
 - **PER** (Person): “Marie Curie”
 - **LOC** (Location): “New York City”
 - **ORG** (Organization): “Stanford University”
 - **GPE** (Geo-Political Entity): “Boulder, Colorado”
- Often multi-word phrases
- But the term is also extended to things that aren't entities:
 - dates, times, prices

Named Entity tagging

The task of named entity recognition (NER):

- find spans of text that constitute proper names
- tag the type of the entity.

NER output

Citing high fuel prices, [ORG **United Airlines**] said [TIME **Friday**] it has increased fares by [MONEY **\$6**] per round trip on flights to some cities also served by lower-cost carriers. [ORG **American Airlines**], a unit of [ORG **AMR Corp.**], immediately matched the move, spokesman [PER **Tim Wagner**] said. [ORG **United**], a unit of [ORG **UAL Corp.**], said the increase took effect [TIME **Thursday**] and applies to most routes where it competes against discount carriers, such as [LOC **Chicago**] to [LOC **Dallas**] and [LOC **Denver**] to [LOC **San Francisco**].

Why NER?

Sentiment analysis: consumer's sentiment toward a particular company or person?

Question Answering: answer questions about an entity?

Information Extraction: Extracting facts about entities from text.

Why NER is hard

1) Segmentation

- In POS tagging, no segmentation problem since each word gets one tag.
- In NER we have to find and segment the entities!

2) Type ambiguity

[PER Washington] was born into slavery on the farm of James Burroughs.
[ORG Washington] went up 2 games to 1 in the four-game series.
Blair arrived in [LOC Washington] for what may well be his last state visit.
In June, [GPE Washington] passed a primary seatbelt law.

BIO Tagging

How can we turn this structured problem into a sequence problem like POS tagging, with one label per word?

[PER Jane Villanueva] of [ORG United] , a unit of [ORG United Airlines Holding] , said the fare applies to the [LOC Chicago] route.

BIO Tagging

[PER Jane Villanueva] of [ORG United] , a unit of [ORG United Airlines Holding] ,
said the fare applies to the [LOC Chicago] route.

Words	BIO Label
Jane	B-PER
Villanueva	I-PER
of	O
United	B-ORG
Airlines	I-ORG
Holding	I-ORG
discussed	O
the	O
Chicago	B-LOC
route	O
.	O

Now we have one tag per token!!!

BIO Tagging

B: token that *begins* a span

I: tokens *inside* a span

O: tokens outside of any span

of tags (where n is #entity types):

1 O tag,

n B tags,

n I tags

total of $2n+1$

Words	BIO Label
Jane	B-PER
Villanueva	I-PER
of	O
United	B-ORG
Airlines	I-ORG
Holding	I-ORG
discussed	O
the	O
Chicago	B-LOC
route	O
.	O

BIO Tagging variants: IO and BIOES

[PER Jane Villanueva] of [ORG United] , a unit of [ORG United Airlines Holding] ,
said the fare applies to the [LOC Chicago] route.

Words	IO Label	BIO Label	BIOES Label
Jane	I-PER	B-PER	B-PER
Villanueva	I-PER	I-PER	E-PER
of	O	O	O
United	I-ORG	B-ORG	B-ORG
Airlines	I-ORG	I-ORG	I-ORG
Holding	I-ORG	I-ORG	E-ORG
discussed	O	O	O
the	O	O	O
Chicago	I-LOC	B-LOC	S-LOC
route	O	O	O
.	O	O	O

Standard algorithms for NER

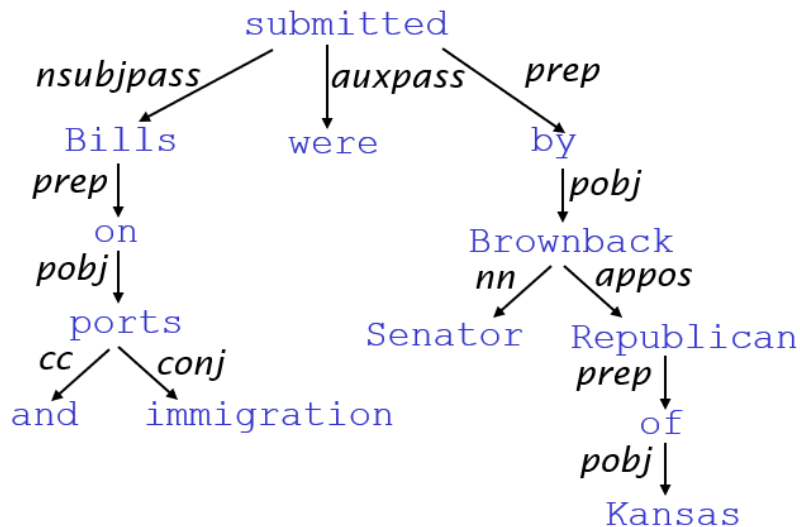
Supervised Machine Learning given a human-labeled training set of text annotated with tags

- Hidden Markov Models
- Conditional Random Fields (CRF)/ Maximum Entropy Markov Models (MEMM)
- Neural sequence models (RNNs or Transformers)
- Large Language Models (like BERT), finetuned
- 2023 and beyond.... Zero-shot prompt to LLM / superseded by end-to-end systems?

Dependency Grammar and Dependency Structure

Dependency syntax postulates that syntactic structure consists of lexical items linked by binary asymmetric relations (“arrows”) called dependencies

The arrows are commonly **typed** with the name of grammatical relations (subject, prepositional object, apposition, etc.)

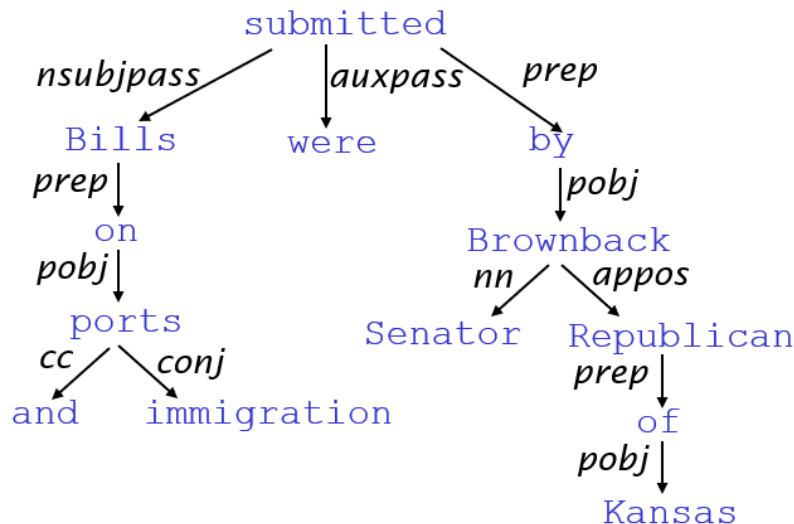


Dependency Grammar and Dependency Structure

Dependency syntax postulates that syntactic structure consists of lexical items linked by binary asymmetric relations (“arrows”) called dependencies

The arrow connects a
head (governor,
superior, regent) with a
dependent (modifier,
inferior, subordinate)

Usually, dependencies
form a tree (connected,
acyclic, single-head)



Relation between phrase structure and dependency structure

- A dependency grammar has a notion of a head. Officially, CFGs don't.
- But modern linguistic theory and all modern statistical parsers (Charniak, Collins, Stanford, ...) do, via hand-written phrasal “head rules”:
 - The head of a Noun Phrase is a noun/number/adj/...
 - The head of a Verb Phrase is a verb/modal/....
- The head rules can be used to extract a dependency parse from a CFG parse

- The closure of dependencies give constituency from a dependency tree
- But the dependents of a word must be at the same level (i.e., “flat”) – there can be no VP!

